MASTER FUNCTIONAL SKILLS

FUNCTIONAL SKILLS MATHEMATICS

CORE REVISION GUIDE • VISUAL METHODS

LEVEL 1

NUMERACY

MATHEMATICS FOUNDATIONS

A step-by-step visual guide to building confidence and passing Level 1 Maths

Level 1 Spec | Emerald Series | Non-Calc Skills | Real-Life Scenarios

MASTER FUNCTIONAL SKILLS · REFORMED MATHEMATICS CURRICULUM DIRECTORY

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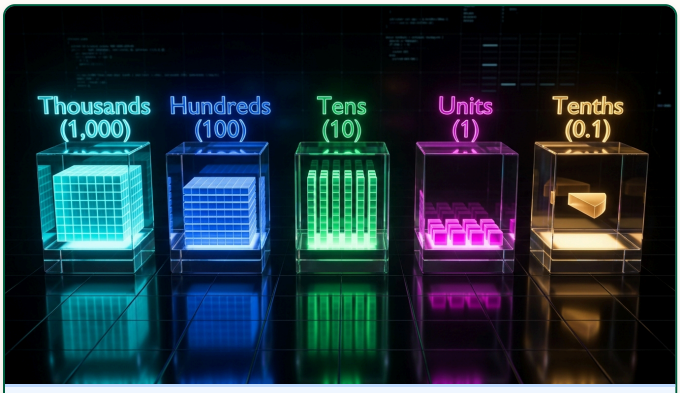
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Master Functional Skills • **Mathematics Manual** • Number, Measures, Shape, Space & Data Handling **Lead Tutor:** Abdul • **Edition:** 2026/2027 Reformed Standards

NUMBER • LEVEL 1

1 Counting, Ordering & Writing Numbers to 1,000,000



Topic 1 Applied Context: Counting, Ordering & Writing Numbe...

Key Method & Rules

Numbers up to 1,000,000 are written with digits grouped in threes. Each position has a distinct place value.

Units, Tens, Hundreds, Thousands, Ten Thousands, Hundred Thousands, Millions.

Step-by-Step Example

Order these warehouse inventory counts from smallest to largest: 48,210; 40,890; 48,090.

1. Compare ten thousands first: all 40,000
2. Compare thousands: 40,890 is smallest
3. Compare hundreds: 48,090 then 48,210
4. Correct order: **40,890; 48,090; 48,210**

Remember!

Always align digits from the right when comparing number size.

Watch Out!

Don't confuse forty thousand with four hundred thousand.

TOPIC 1 • CANDIDATE WORKSPACE

P Counting, Ordering & Writing Numbers to 1,000,000: Practice Drills

1 **Question 1 (Foundation):** Write 'Forty-five thousand, eight hundred and twenty' in figures. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A factory produced 128,450 units in March and 95,200 units in April. How many more units were produced in March? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Put these warehouse stock counts in order from smallest to largest: 14,200; 3,450; 108,500; 12,890. **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

2 Negative Numbers in Practical Contexts



Topic 2 Applied Context: Negative Numbers in Practical Conte...

Key Method & Rules

Negative numbers show values below zero such as freezer temperatures or bank overdrafts.

Temperature rise: count up through zero. Overdraft: negative balance means money owed to the bank.

Step-by-Step Example

A cold storage freezer is at -18°C. A sensor alert triggers if temperature rises by 7°C. What is the new reading?

1. Start at -18°C
2. Add 7°C: -18 + 7
3. New temperature = **-11°C**

Remember!

Remember: moving towards zero increases temperature.

Watch Out!

A smaller negative number is actually warmer than a larger negative number!

TOPIC 2 • CANDIDATE WORKSPACE

P Negative Numbers in Practical Contexts: Practice Drills

1 **Question 1 (Foundation):** Order these temperatures from coldest to warmest: +4°C, −8°C, −2°C, 0°C, −11°C. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A cold store is at −15°C. When the loading door opens, the temperature rises by 6°C. What is the new temperature? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A trader has a bank balance of £120. A bill of £185 is taken from the account. What is the new balance, and how much must be paid in to reach +£50? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

3 Column Addition & Subtraction with Money



Topic 3 Applied Context: Column Addition & Subtraction with ...

★ Key Method & Rules

Align decimal points vertically in straight columns before adding or subtracting monetary values.

Add or subtract from right to left, carrying or borrowing across columns as needed.

✎ Step-by-Step Example

A tradesperson buys supplies: £142.50, £68.75, and £29.99. Find the total cost.

1. Align decimal points: 142.50 + 68.75 + 29.99
2. Sum pennies: 0 + 5 + 9 = 14 (carry 1)
3. Total = **£241.24**

✓ Remember!

Always include the £ sign and exactly two decimal places in money answers.

! Watch Out!

Keep digits in neat vertical columns to avoid alignment slips.

TOPIC 3 • CANDIDATE WORKSPACE

P Column Addition & Subtraction with Money: Practice Drills

1 **Question 1 (Foundation):** Calculate without a calculator: 4,850 + 2,680. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A tradesperson buys three tool accessories costing £24.50, £18.75, and £9.20. Calculate the total cost. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A department has a monthly stationery budget of £1,500. They spend £845 on paper and £320 on toner cartridges. How much budget remains? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

4 Multiplication of 2-Digit & 3-Digit Numbers



Topic 4 Applied Context: Multiplication of 2-Digit & 3-Digit Nu...

★ Key Method & Rules

Long multiplication allows multiplying multi-digit quantities in vocational purchasing tasks.

Multiply by units first, then multiply by tens (remember the zero placeholder!), then add.

✎ Step-by-Step Example

A clinic orders 24 boxes of medical gloves. Each box contains 125 pairs. How many pairs in total?

1. Multiply 125 x 4 = 500
2. Multiply 125 x 20 = 2,500 (placeholder 0)
3. Add: 500 + 2,500 = **3,000 pairs**

✓ Remember!

Never forget the zero in the second row when multiplying by the tens digit.

! Watch Out!

Estimate first: 25 x 120 = 3,000 to verify your answer.

P

TOPIC 4 • CANDIDATE WORKSPACE

Multiplication of 2-Digit & 3-Digit Numbers: Practice Drills

1 **Question 1 (Foundation):** Calculate without using a calculator: 145 × 18. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A warehouse receives 24 crates of parts. Each crate holds 35 items. How many items are delivered in total? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An operative works 32 hours in a week at an hourly rate of £12.50. Calculate their total weekly gross pay. **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

5 Division with Remainders in Context



Topic 5 Applied Context: Division with Remainders in Context

★ Key Method & Rules

Division splits quantities into equal shares. Interpret remainders based on the real-world situation.

Round UP if packing items into boxes; round DOWN if counting complete sets.

✎ Step-by-Step Example

An operative has 148 components. Each shipping crate holds 12 components. How many crates are needed?

1. Divide: 148 / 12 = 12 remainder 4
2. 12 crates will leave 4 components unpacked
3. Total crates needed to ship all items = **13 crates**

✓ Remember!

Always interpret the remainder: does the scenario need an extra crate or van?

! Watch Out!

Don't leave an exam answer as '12 r 4' when asked for full crates.

TOPIC 5 • CANDIDATE WORKSPACE

P

Division with Remainders in Context: Practice Drills

1 **Question 1 (Foundation):** Calculate without a calculator: 256 ÷ 8. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A dispatch team must pack 185 components into crates that hold 12 components each. How many crates are needed to pack all 185 components? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Four staff members share a £350 team bonus equally. How much does each person receive? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

6 Fractions of Quantities: Unit & Non-Unit



Topic 6 Applied Context: Fractions of Quantities: Unit & Non-U...

Key Method & Rules

To find a fraction of an amount: Divide by the bottom (denominator), then multiply by the top (numerator).

Amount / Denominator x Numerator = Fraction of amount.

Step-by-Step Example

Find 3/4 of £360 for an equipment deposit payment.

1. Divide by 4: £360 / 4 = £90 (one quarter)
2. Multiply by 3: £90 x 3 = **£270**

Remember!

Remember: 'Divide by the bottom, times by the top'.

Watch Out!

Check your answer is smaller than the original amount for proper fractions.

TOPIC 6 • CANDIDATE WORKSPACE

P

Fractions of Quantities: Unit & Non-Unit: Practice Drills

1 **Question 1 (Foundation):** Calculate 1/4 of £240. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A building load weighs 480 kg. **(2 marks)** A crane lifts 3/4 of this weight in the first load. How many kilograms does it lift?

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A training session lasts 90 minutes. 2/3 of the session is spent on practical tasks. How many minutes are spent on practical tasks? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

7 Decimals: Ordering, Place Value & Rounding



Topic 7 Applied Context: Decimals: Ordering, Place Value & Ro...

★ Key Method & Rules

Decimals represent parts of a whole: tenths (0.1), hundredths (0.01), and thousandths (0.001).

To round to 1 decimal place: look at the hundredths digit. 5 or more rounds up; 4 or less stays.

✎ Step-by-Step Example

Round 14.86 metres to 1 decimal place for a carpet fitting measurement.

1. Look at second decimal place: 6 (hundredths)
2. Since 6 >= 5, round the tenths up from 8 to 9
3. Rounded = **14.9 metres**

✓ Remember!

Add trailing zeros when aligning decimals of different lengths (e.g. 3.40 and 3.25).

! Watch Out!

Money is always rounded to 2 decimal places.

TOPIC 7 • CANDIDATE WORKSPACE

P Decimals: Ordering, Place Value & Rounding: Practice Drills

1 **Question 1 (Foundation):** Round 14.86 metres to 1 decimal place. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A customer buys items costing £3.45 and £8.80. Calculate the total bill. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A joiner cuts a 1.85 m piece from a 4.2 m timber batten. What length of batten is left in metres? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

8 Simple Percentages: 10%, 25%, 50% & 75%



Topic 8 Applied Context: Simple Percentages: 10%, 25%, 50% ...

Key Method & Rules

Key percentage fractions: 50% = half (/ 2); 25% = quarter (/ 4); 10% = divide by 10; 75% = 3/4.

Build other percentages from 10%: 20% = 10% x 2; 5% = half of 10%; 15% = 10% + 5%.

Step-by-Step Example

A sofa costs £480. In a seasonal sale it is reduced by 25%. Find the sale price.

1. Find 25% (divide by 4): £480 / 4 = £120 discount
2. Deduct discount: £480 - £120
3. Sale price = **£360.00**

Remember!

Always read carefully: does the question ask for the DISCOUNT or the SALE PRICE?

Watch Out!

10% is the master building block — find 10% by shifting digits one place right.

TOPIC 8 • CANDIDATE WORKSPACE

P Simple Percentages: 10%, 25%, 50% & 75%: Practice Drills

1 **Question 1 (Foundation):** Calculate 10% of £450. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A jacket priced at £180 is reduced by 25% in a clearance sale. Calculate the new sale price. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A company employs 320 staff. 75% of the staff completed fire safety training this week. How many staff completed the training? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

9 Simple Ratio: Sharing in 2 Parts



Topic 9 Applied Context: Simple Ratio: Sharing in 2 Parts

★ Key Method & Rules

Ratios compare parts of a whole. To share in a ratio, find the value of one part.

Add parts -> Total / Parts = 1 part -> Multiply by each ratio number.

✎ Step-by-Step Example

Share £240 between two charity branches in the ratio 3 : 1.

1. Total parts: 3 + 1 = 4 parts
2. One part: £240 / 4 = £60
3. Branch 1 (3 parts): 3 x £60 = £180
4. Branch 2 (1 part): 1 x £60 = **£60**

✓ Remember!

Always check the two shares add back up to the starting total (£180 + £60 = £240).

! Watch Out!

Keep the numbers in the same order as written in the question.

TOPIC 9 • CANDIDATE WORKSPACE

P

Simple Ratio: Sharing in 2 Parts: Practice Drills

1 **Question 1 (Foundation):** Share £180 between two charities in the ratio 2 : 1. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A builder mixes cement and sand in the ratio 1 : 4. How much cement and how much sand are in a 25 kg bag of mix? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A decorator mixes white and blue paint in the ratio 2 : 1 to make 6 litres of light blue paint. How many litres of each colour are used? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

10 Simple Proportion & Scaling Recipes



Topic 10 Applied Context: Simple Proportion & Scaling Recipes

Key Method & Rules

Direct proportion means if you double one amount, you double the other.

Find the quantity for 1 portion/item (unitary method), then multiply by the target number.

Step-by-Step Example

A soup recipe for 4 people needs 600 ml of stock. How much stock is needed for 6 people?

1. Stock for 1 person: 600 ml / 4 = 150 ml
2. Stock for 6 people: 150 ml x 6 = **900 ml**

Remember!

Divide by the starting number of people first, then multiply by the new number.

Watch Out!

Check units match: convert ml to litres if required.

TOPIC 10 • CANDIDATE WORKSPACE

P

Simple Proportion & Scaling Recipes: Practice Drills

1 **Question 1 (Foundation):** A soup recipe for 4 people uses 600 ml of vegetable stock. How much stock is needed to make the soup for 6 people? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Three identical training manuals cost £24.00. How much do five training manuals cost? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A cake recipe for 8 portions needs 250 g of flour and 150 g of sugar. How much flour and sugar are needed for 16 portions? **(3 marks)**

Show your working

Answer: _____

NUMBER • LEVEL 1

11 Estimation & Checking Calculations



Topic 11 Applied Context: Estimation & Checking Calculations

★ Key Method & Rules

Estimation checks whether an answer is sensible by rounding numbers to the nearest 10, 100, or whole number.

Round numbers before calculating: 49 x 21 ~ 50 x 20 = 1,000.

✎ Step-by-Step Example

Estimate the total cost of 19 safety helmets at £11.85 each.

1. Round 19 to 20
2. Round £11.85 to £10 or £12
3. Estimate: 20 x £10 = £200 (or 20 x £12 = **£240**)

✓ Remember!

Examiners award separate marks for showing an estimated check of your answer.

! Watch Out!

Always write down the rounded calculation clearly.

P

TOPIC 11 • CANDIDATE WORKSPACE

Estimation & Checking Calculations: Practice Drills

1 **Question 1 (Foundation):** Round each number to the nearest 10 to estimate: 49 × 21. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Estimate the total cost of 19 high-visibility vests at £11.85 each by rounding both numbers. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A warehouse had 500 boxes and dispatched 184 boxes, leaving 316 boxes. Write down a reverse addition calculation to check that 316 is correct. **(3 marks)**

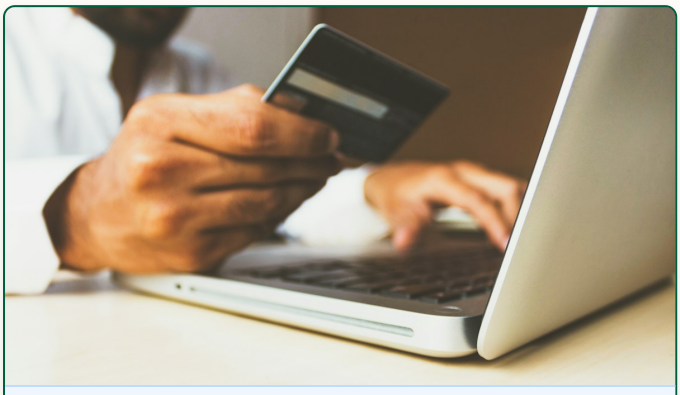
Show your working

Answer: _____

12

MEASURE • LEVEL 1

Money & Giving Change



Topic 12 Applied Context: Money & Giving Change

Key Method & Rules

Calculate total spend and work out change from £10, £20, or £50 notes.

Change = Amount given - Total spend.

Step-by-Step Example

A customer buys items costing £3.45, £2.80, and £6.15. They pay with a £20 note. Calculate their change.

1. Total cost: 3.45 + 2.80 + 6.15 = £12.40
2. Calculate change: £20.00 - £12.40
3. Change = **£7.60**

Remember!

Line up decimal points when calculating change.

Watch Out!

Count up from the bill to the payment note to double-check.

TOPIC 12 • CANDIDATE WORKSPACE

P

Money & Giving Change: Practice Drills

1 **Question 1 (Foundation):** A shopper buys items costing £3.45, £2.80, and £6.15. How much change do they receive from a £20 note? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A customer pays for a £34.20 meal with a £50 note. Calculate the change due. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An apprentice buys 3 packs of drill bits at £4.50 per pack and pays with a £20 note. Calculate the total cost and the change given. **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

13 12-Hour & 24-Hour Clock Time



Topic 13 Applied Context: 12-Hour & 24-Hour Clock Time

Key Method & Rules

24-hour time uses 4 digits from 00:00 to 23:59 without am or pm. For times after 12:00, add 12 to the pm hour.

To convert pm to 24h: add 12 to hour (3:15 pm = 15:15). To convert 24h to pm: subtract 12 (19:40 = 7:40 pm).

Step-by-Step Example

A shift starts at 08:30 and finishes at 16:45. How long was the shift?

1. Count hours from 08:30 to 16:30 = 8 hours
2. Count remaining minutes from 16:30 to 16:45 = 15 minutes
3. Total duration = **8 hours 15 minutes**

Remember!

There are 60 minutes in an hour, NOT 100! 8 hours 15 mins is not 8.15 hours.

Watch Out!

00:00 is midnight; 12:00 is midday.

TOPIC 13 • CANDIDATE WORKSPACE

P

12-Hour & 24-Hour Clock Time: Practice Drills

1 **Question 1 (Foundation):** Convert: a) 3:15 pm to 24-hour clock b) 19:40 to 12-hour am/pm clock. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A hospital shift starts at 08:30 and ends at 16:45. How long is the shift in hours and minutes? **(2 marks)**

Show your working

Answer: _____

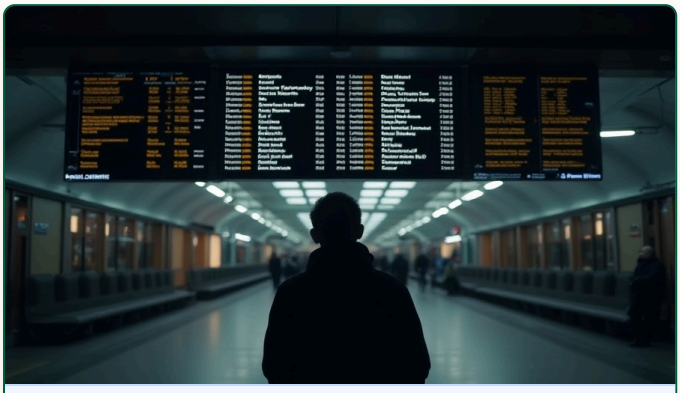
3 **Question 3 (Exam Challenge):** A staff training meeting starts at 14:30 and lasts for 1 hour 45 minutes. At what time does the meeting finish? **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

14 Reading Bus & Train Timetables



Topic 14 Applied Context: Reading Bus & Train Timetables

Key Method & Rules

Timetables show departure and arrival times in columns or rows.

Read down a column to track a single vehicle journey; read across a row to see departures from one stop.

Step-by-Step Example

A bus leaves High Street at 14:22 and arrives at the Hospital at 14:58. How long does the journey take?

1. From 14:22 to 14:58
2. Calculate difference: 58 - 22 minutes
3. Journey time = **36 minutes**

Remember!

Check for notes like 'Saturdays only' or 'Does not run on Bank Holidays'.

Watch Out!

Calculate journey duration by counting up to the next hour first.

TOPIC 14 • CANDIDATE WORKSPACE

P

Reading Bus & Train Timetables: Practice Drills

1 **Question 1 (Foundation):** A bus departs High Street at 14:22 and arrives at the Hospital at 14:58. How many minutes does the journey take? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Trains leave Station A at 10:05, 10:32, and 11:02, taking 33 minutes to reach Station B. If you arrive at Station A at 10:15, when will you reach Station B? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A coach leaves London at 09:45 and arrives in Bristol at 11:03. Calculate the total journey time in hours and minutes. **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

15 Metric Units of Length: mm, cm, m, km



Topic 15 Applied Context: Metric Units of Length: mm, cm, m, ...

Key Method & Rules

Metric length conversions: 10 mm = 1 cm; 100 cm = 1 m; 1,000 m = 1 km.

To convert smaller unit to larger: divide.
To convert larger unit to smaller: multiply.

Step-by-Step Example

A wooden skirting board is 2.4 metres long. Convert this length into centimetres.

1. Conversion: 1 metre = 100 centimetres
2. Calculation: 2.4 x 100
3. Length = **240 cm**

Remember!

Multiply by 100 to convert metres to cm; divide by 100 to convert cm to metres.

Watch Out!

Remember: mm is millimetres, cm is centimetres, m is metres, km is kilometres.

TOPIC 15 • CANDIDATE WORKSPACE

P

Metric Units of Length: mm, cm, m, km: Practice Drills

1 **Question 1 (Foundation):** Convert: a) 2.4 metres into centimetres b) 45 millimetres into centimetres. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** From a 3.6 m length of timber, a carpenter cuts 4 pieces each measuring 65 cm. What length of timber remains in metres? **(2 marks)**

Show your working

Answer: _____

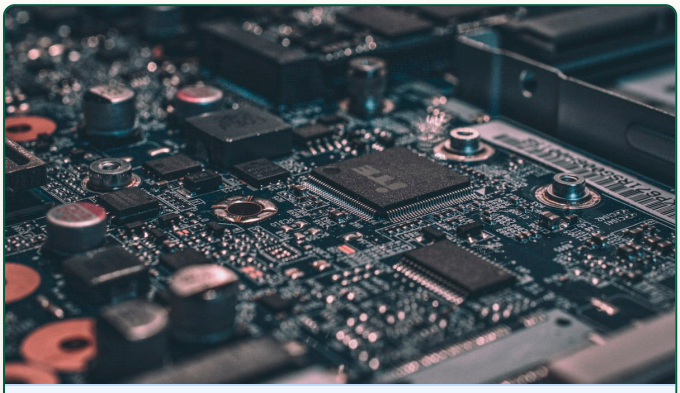
3 **Question 3 (Exam Challenge):** A charity runner completes a 5.2 km run. How many metres is this in total? **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

16 Metric Units of Weight: g and kg



Topic 16 Applied Context: Metric Units of Weight: g and kg

★ Key Method & Rules

Metric mass conversions: 1,000 grams (g) = 1 kilogram (kg).

Grams to kg: divide by 1,000. Kg to grams: multiply by 1,000.

✎ Step-by-Step Example

A parcel weighs 1,750 grams. What is its weight in kilograms?

1. Conversion: 1 kg = 1,000 g
2. Calculation: 1,750 / 1,000
3. Weight = **1.75 kg**

✓ Remember!

Divide by 1,000 by moving digits three places to the right.

! Watch Out!

A bag of sugar weighs approximately 1 kg.

P

TOPIC 16 • CANDIDATE WORKSPACE

Metric Units of Weight: g and kg: Practice Drills

1 **Question 1 (Foundation):** Convert 1,750 grams into kilograms. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A courier packs 4 parcels, each weighing 350 grams. What is the total weight of the 4 parcels in kilograms? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A chef has a 2.5 kg bag of flour and uses 800 grams for a batch of bread. How many kilograms of flour remain in the bag? **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

17 Metric Units of Capacity: ml and litres



Topic 17 Applied Context: Metric Units of Capacity: ml and litr...

★ Key Method & Rules

Liquid capacity conversions: 1,000 millilitres (ml) = 1 litre (l).

Ml to litres: divide by 1,000. Litres to ml: multiply by 1,000.

✎ Step-by-Step Example

A cleaning operative mixes 250 ml of disinfectant into 4.5 litres of water. Convert 4.5 litres to ml.

1. Conversion: 1 litre = 1,000 ml
2. Calculation: 4.5 x 1,000
3. Water volume = **4,500 ml**

✓ Remember!

Always convert all quantities to the SAME unit before adding or comparing.

! Watch Out!

1 litre of water weighs exactly 1 kilogram.

TOPIC 17 • CANDIDATE WORKSPACE

P

Metric Units of Capacity: ml and litres: Practice Drills

1 **Question 1 (Foundation):** Convert 4.5 litres into millilitres. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A catering urn dispenses 15 cups of tea, each holding 250 ml. How many litres of tea were dispensed altogether? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A cleaner mixes 500 ml of disinfectant concentrate with 4.5 litres of water. What is the total volume of the diluted solution in litres? **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

18 Perimeter of Rectangles & Simple Shapes



Topic 18 Applied Context: Perimeter of Rectangles & Simple S...

★ Key Method & Rules

Perimeter is the total boundary distance around the outside of a 2D shape.

Rectangle perimeter = 2 x (length + width) or sum of all 4 sides.

✎ Step-by-Step Example

A rectangular garden is 9 metres long and 6 metres wide. Find the total fencing needed to enclose it.

1. Formula: 2 x (length + width)
2. Calculation: 2 x (9 + 6) = 2 x 15
3. Perimeter = **30 metres**

✓ Remember!

Don't forget to include all 4 sides of a rectangle, not just the two labelled sides.

! Watch Out!

Perimeter is measured in simple units (m, cm), NOT square units.

TOPIC 18 • CANDIDATE WORKSPACE

P

Perimeter of Rectangles & Simple Shapes: Practice Drills

1 **Question 1 (Foundation):** Calculate the perimeter of a rectangular garden measuring 9 metres long by 6 metres wide. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A rectangular bedroom measures 4.5 m by 3.2 m. Calculate the total perimeter of the room. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A rectangular car park measures 15 metres long by 9 metres wide. Calculate the total length of boundary fencing needed. **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

19 Area of Rectangles: Length x Width



Topic 19 Applied Context: Area of Rectangles: Length x Width

★ Key Method & Rules

Area is the flat surface space inside a 2D shape, measured in square units (m², cm²).

Area of a rectangle = length x width.

✎ Step-by-Step Example

A bedroom floor is 4.5 metres long and 3 metres wide. What is the area of carpet needed?

1. Formula: Area = length x width
2. Calculation: 4.5 x 3
3. Area = **13.5 m²**

✓ Remember!

Area answers MUST have square unit labels: m², cm², or mm².

! Watch Out!

Do not confuse area (length x width) with perimeter (distance around edge).

TOPIC 19 • CANDIDATE WORKSPACE

P

Area of Rectangles: Length x Width: Practice Drills

1 **Question 1 (Foundation):** Calculate the area of a floor measuring 4.5 metres long by 3 metres wide. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A wall measures 6 metres long by 2.5 metres high. Calculate its area, and find the total area to be painted if 2 coats are required. **(2 marks)**

Show your working

Answer: _____

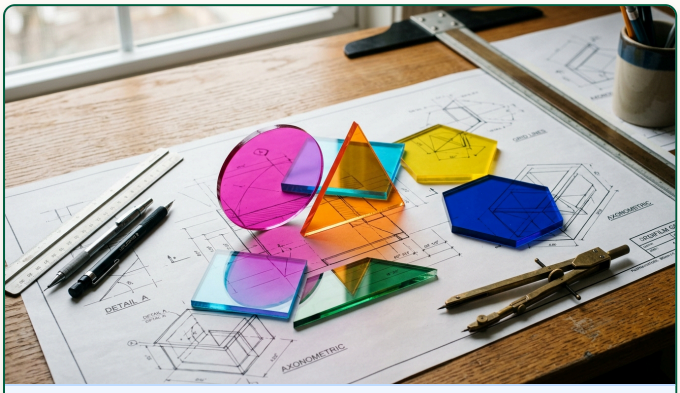
3 **Question 3 (Exam Challenge):** A rectangular office measures 8 metres by 5 metres. Carpet tiles cost £12 per square metre. Calculate the floor area and total cost. **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

20 Recognising 2D & 3D Shapes & Angles



Topic 20 Applied Context: Recognising 2D & 3D Shapes & Angl...

Key Method & Rules

2D shapes: square, rectangle, triangle, circle, pentagon, hexagon. 3D shapes: cube, cuboid, cylinder, sphere, cone, pyramid.

Right angle = 90° (square corner).
Straight line = 180°. Full turn = 360°.

Step-by-Step Example

Identify the 3D shape that has 6 rectangular faces, 12 edges, and 8 vertices.

1. 6 rectangular faces = **Cuboid** (rectangular prism)
2. A cube has 6 square faces; a cuboid has rectangular faces

Remember!

Vertices are corners; edges are straight boundary lines; faces are flat surfaces.

Watch Out!

A right angle is indicated with a small square in the corner.

P

TOPIC 20 • CANDIDATE WORKSPACE

Recognising 2D & 3D Shapes & Angles: Practice Drills

1 **Question 1 (Foundation):** Name the 3D shape that has 6 rectangular faces, 12 edges, and 8 vertices. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** How many degrees are in a right angle and in a straight line? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** State the number of faces, edges, and vertices on a triangular prism. **(3 marks)**

Show your working

Answer: _____

MEASURE • LEVEL 1

21 Compass Directions & Bearings



Topic 21 Applied Context: Compass Directions & Bearings

Key Method & Rules

The 8 main compass points: North, North-East, East, South-East, South, South-West, West, North-West.

Remember: 'Naughty Elephants Squirt Water' for North, East, South, West (clockwise).

Step-by-Step Example

A delivery van is travelling North and turns 90° clockwise. Which direction is it now facing?

1. Start facing North
2. Turn 90° clockwise = one quarter turn right
3. New heading = **East**

Remember!

Clockwise is the direction clock hands move; anti-clockwise is to the left.

Watch Out!

Between North and East is North-East (45°).

TOPIC 21 • CANDIDATE WORKSPACE

P

Compass Directions & Bearings: Practice Drills

1 **Question 1 (Foundation):** You are facing North and turn 90° clockwise. Which compass direction are you now facing? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Which compass direction is directly opposite South-West? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** A delivery driver is heading East and makes a 180° turn. Which direction is the driver now travelling? **(3 marks)**

Show your working

Answer: _____

DATA • LEVEL 1

22 Tally Charts & Frequency Tables



Topic 22 Applied Context: Tally Charts & Frequency Tables

★ Key Method & Rules

Tally marks record counts in groups of five (four vertical lines with a diagonal cross).

Count tallies to find the frequency (total count) for each category.

✎ Step-by-Step Example

A transport survey records 14 tallies for car travel. Write the tally representation.

1. Two complete bundles of five = 10 tallies
2. Four single tally lines = 4 tallies
3. Total = **5 + 5 + 4 = 14**

✓ Remember!

Every fifth mark crosses through the group of four.

! Watch Out!

Always check the sum of frequencies equals the total sample size.

TOPIC 22 • CANDIDATE WORKSPACE

P

Tally Charts & Frequency Tables: Practice Drills

1 **Question 1 (Foundation):** Explain how 14 is written in tally marks using bundles of five. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A traffic count records: Car (8), Van (14), Bus (12), Bike (6). Calculate the total frequency of vehicles. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** From the traffic count in Question 2, which vehicle category is the most popular (the mode)? **(3 marks)**

Show your working

Answer: _____

DATA • LEVEL 1

23 Bar Charts: Scales & Comparison



Topic 23 Applied Context: Bar Charts: Scales & Comparison

Key Method & Rules

Bar charts represent categories with rectangular bars of equal width. The height shows the frequency.

Read the vertical axis scale carefully: check what each grid line represents (1, 2, 5, 10).

Step-by-Step Example

In a bar chart, Monday has 25 sales and Wednesday has 40 sales. How many more sales were made on Wednesday?

1. Read values: Wed = 40, Mon = 25
2. Subtract: 40 - 25
3. Difference = **15 sales**

Remember!

Leave equal gaps between bars in a categorical bar chart.

Watch Out!

Always check what one square on the scale is worth before reading values.

P

TOPIC 23 • CANDIDATE WORKSPACE

Bar Charts: Scales & Comparison: Practice Drills

1 **Question 1 (Foundation):** A bar chart shows 25 sales on Monday and 40 sales on Wednesday. How many more sales were made on Wednesday? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** On a bar chart axis, 4 grid squares represent 20 units. How many units does 1 grid square represent? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Daily sales from Monday to Friday are: 25, 30, 40, 35, 35. Calculate the total weekly sales. **(3 marks)**

Show your working

Answer: _____

24

DATA • LEVEL 1

Pictograms with Key Symbols



Topic 24 Applied Context: Pictograms with Key Symbols

Key Method & Rules

Pictograms use picture symbols to show data. A key explains what each full symbol represents.

Part of a symbol represents a fraction of the value (e.g. half a circle = half the key value).

Step-by-Step Example

A pictogram key shows 1 circle = 8 vehicles. How many vehicles are represented by 3.5 circles?

1. 3 full circles: 3 x 8 = 24
2. Half circle: 0.5 x 8 = 4
3. Total = 24 + 4 = **28 vehicles**

Remember!

Always check the KEY first — never assume 1 symbol represents 1 item!

Watch Out!

Half a symbol represents half of the key's value.

TOPIC 24 • CANDIDATE WORKSPACE

P

Pictograms with Key Symbols: Practice Drills

1 **Question 1 (Foundation):** A pictogram key shows 1 circle = 8 cars. How many cars are represented by 3.5 circles? **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** If 1 symbol represents 4 bicycles, how many symbols are needed to show 20 bicycles? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** In the car pictogram (1 circle = 8 cars), Monday has 3 circles, Tuesday has 2.5 circles, and Wednesday has 3.5 circles. Find the total cars across all three days. **(3 marks)**

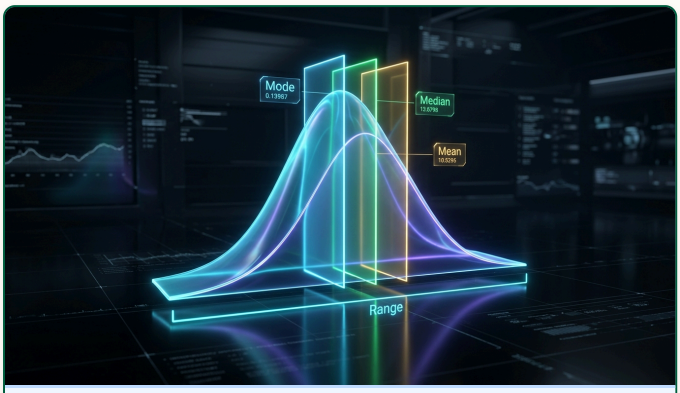
Show your working

Answer: _____

25

DATA • LEVEL 1

Mean Average: Total / Count



Topic 25 Applied Context: Mean Average: Total / Count

Key Method & Rules

The mean is the average found by adding all values together, then dividing by the number of values.

Mean = Sum of all values / Number of values.

Step-by-Step Example

A worker's daily hours over 5 days are: 7, 8, 8, 9, 8. Find their mean daily working hours.

1. Sum all values: 7 + 8 + 8 + 9 + 8 = 40 hours
2. Divide by 5 days: 40 / 5
3. Mean daily hours = **8 hours**

Remember!

The mean must be between the smallest and largest values in the data set.

Watch Out!

If the mean doesn't divide evenly, round sensibly (e.g. to 1 decimal place).

TOPIC 25 • CANDIDATE WORKSPACE

P

Mean Average: Total / Count: Practice Drills

1 **Question 1 (Foundation):** An employee works 7, 8, 8, 9, and 8 hours over 5 days. Calculate the mean daily hours worked. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Four learners score 15, 18, 21, and 26 on a safety test. Calculate the mean score. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Over 4 weeks, a household spends £40, £45, £38, and £47 on electricity. Calculate the mean weekly spend. **(3 marks)**

Show your working

Answer: _____

DATA • LEVEL 1

26 Range: Measure of Spread (High - Low)



Topic 26 Applied Context: Range: Measure of Spread (High - L...

★ Key Method & Rules

The range measures how spread out data is: Range = Highest value - Lowest value.

A smaller range means results are more consistent; a larger range means more varied.

✎ Step-by-Step Example

A department records daily call volumes: 45, 62, 38, 55, 71, 49. Find the range of calls.

1. Identify highest value = 71
2. Identify lowest value = 38
3. Range: 71 - 38 = **33 calls**

✓ Remember!

Range is NOT an average — it is a measure of consistency or variation.

! Watch Out!

Scan the list carefully to ensure you pick the true maximum and minimum.

TOPIC 26 • CANDIDATE WORKSPACE

P

Range: Measure of Spread (High - Low): Practice Drills

1 **Question 1 (Foundation):** Daily call counts in an office are: 45, 62, 38, 55, 71, 49. Calculate the range of calls. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Overnight temperatures were −4°C on Monday and +6°C on Friday. What is the temperature range? **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Delivery times for 5 parcels were 18, 29, 45, 31, and 24 minutes. Calculate the range of the delivery times. **(3 marks)**

Show your working

Answer: _____

DATA • LEVEL 1

27 Reading Information from Data Tables



Topic 27 Applied Context: Reading Information from Data Tab...

★ Key Method & Rules

Tables present information organized into labelled rows and columns.

Find the required row and column, and read the value where they intersect.

✎ Step-by-Step Example

A price table shows postage costs: Letter (£0.85), Large Letter (£1.55), Small Parcel (£3.29). Find cost for 3 parcels.

1. Locate Small Parcel row = £3.29
2. Multiply by 3: 3 x £3.29
3. Total cost = **£9.87**

✓ Remember!

Read table headers and footnote conditions carefully.

! Watch Out!

Cross-reference row and column headers with a ruler if needed.

TOPIC 27 • CANDIDATE WORKSPACE

P

Reading Information from Data Tables: Practice Drills

1 **Question 1 (Foundation):** A postage table shows: Letter £0.85, Large Letter £1.55, Small Parcel £3.29. Find the cost of sending 3 Small Parcels. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** A courier tariff table shows Standard Delivery (£4.50) and Express Next-Day (£8.95) for orders under £50. State the Standard Delivery charge. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** An invoice table lists: 4 chairs at £28.50 each and 1 table at £34.20. Calculate the total invoice amount. **(3 marks)**

Show your working

Answer: _____

DATA • LEVEL 1

28 Probability: Certain, Likely, Evens, Unlikely, Impossible



Topic 28 Applied Context: Probability: Certain, Likely, Evens, ...

★ Key Method & Rules

Probability describes how likely an event is using words: Impossible (0), Unlikely, Evens (0.5 / 50-50), Likely, Certain (1).

Evens means equal chance of happening or not happening (e.g. flipping a coin).

✎ Step-by-Step Example

What is the probability of rolling a number between 1 and 6 on a standard fair dice?

1. Every face is between 1 and 6
2. Outcome is guaranteed: **Certain** (Probability = 1)

✓ Remember!

Evens is 50-50 (half chance).

! Watch Out!

Impossible means there is zero chance of the event occurring.

TOPIC 28 • CANDIDATE WORKSPACE

P Probability: Certain, Likely, Evens, Unlikely, Impossible: Practice Drills

1 **Question 1 (Foundation):** Use a probability word (Impossible, Unlikely, Evens, Likely, Certain) to describe rolling a number from 1 to 6 on a standard fair dice. **(2 marks)**

Show your working

Answer: _____

2 **Question 2 (Workplace Problem):** Describe the likelihood of flipping tails on a fair 50p coin. **(2 marks)**

Show your working

Answer: _____

3 **Question 3 (Exam Challenge):** Describe the likelihood of rolling an 8 on a standard 6-sided dice numbered 1 to 6. **(3 marks)**

Show your working

Answer: _____

CP1

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 1 • Part 1/4

1 **Question 1:** Calculate without using a calculator: 348 × 16. Show your full written method. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account is overdrawn at -£185. The customer pays in £450. What is the new balance? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Simplify fully: a) 18/24 b) Write 3/5 as a percentage. **(2 marks)**

Show your working

Answer: _____

CP1

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 1 • Part 2/4

3 **Question 4:** Work out: 1,476 ÷ 6 without a calculator. Show your bus stop division. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** Evaluate: 20 - 4 × 3 + 2. **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** Share £120 in the ratio 3 : 1. What is the larger share? **(2 marks)**

Show your working

Answer: _____

CP1

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 1 • Part 3/4

5 **Question 7:** A jacket costs £80. It is reduced by 15% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A recipe for 4 people requires 300 g of rice. How much rice is needed for 9 people? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A delivery company has 350 parcels to deliver. In the morning they deliver 3/5 of the parcels. How many parcels remain to be delivered in the afternoon? **(3 marks)**

Show your working

Answer: _____

CP1

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 1 • Part 4/4

7 **Question 10:** An electrician charges: £45 call-out + (£32 × hours). He works for 4.5 hours. How much is the bill? **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A supermarket sells a 6-pack of cola for £4.20 and a 10-pack of the same cola for £6.80. Which pack is better value per can? Show your calculations. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A landscape gardener orders 24 bags of decorative gravel at £6.75 each. Delivery is £25.00. He has a budget of £180.00. Does he have enough money? **(3 marks)**

Show your working

Answer: _____

LEVEL 1 REVIEW • EXAM STYLE

CP2

Checkpoint Assessment 2 • Part 1/4

1 **Question 1:** Convert: a) 4.2 kg to grams b) 650 ml to litres. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A train departs at 09:35 and arrives at 11:15. How long was the journey in hours and minutes? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** Calculate the perimeter of a rectangle measuring 8 m by 5.5 m. **(2 marks)**

Show your working

Answer: _____

LEVEL 1 REVIEW • EXAM STYLE

CP2

Checkpoint Assessment 2 • Part 2/4

3 **Question 4:** Calculate the area of a square patio with side length 6 m. **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A bottle of medicine holds 250 ml. A patient takes 15 ml per dose. How many full doses can be taken? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** State the number of faces, edges, and vertices on a cuboid. **(2 marks)**

Show your working

Answer: _____

CP2

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 2 • Part 3/4

5 **Question 7:** A carpet costs £18.50 per square metre. A living room measures 5.4 m by 4.0 m. Calculate the cost of the carpet. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** A delivery driver travels 30 miles. Using 5 miles = 8 km, convert this distance into kilometres. **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A water cistern measures 1.2 m long, 0.8 m wide, and 0.6 m high. How many litres of water does it hold? (1 m^3 = 1000 L) **(3 marks)**

Show your working

Answer: _____

LEVEL 1 REVIEW • EXAM STYLE

CP2

Checkpoint Assessment 2 • Part 4/4

7 **Question 10:** A healthcare assistant works 32 basic hours at £11.20/hr, plus 5 hours overtime at time-and-a-half (£16.80/hr). What is her total pay? **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** On a map with scale 1 cm = 50 m, the distance between two bus stops is 7.4 cm. What is the real distance in metres? **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** An L-shaped lawn has two sections: 6 m × 4 m and 3 m × 2 m. Grass seed costs £4.50 per box. Each box covers 10 m². How much does it cost to seed the lawn? **(3 marks)**

Show your working

Answer: _____

CP3

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 3 • Part 1/4

1 **Question 1:** Test scores: 12, 18, 15, 11, 14. Calculate: a) Mean b) Range. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** A spinner has 6 equal sections numbered 1 to 6. What is the probability of spinning an odd number? **(2 marks)**

Show your working

Answer: _____

3 **Question 3:** A tally chart records: 𝍸 𝍸 |||. Write the frequency number. **(2 marks)**

Show your working

Answer: _____

CP3

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 3 • Part 2/4

3 **Question 4:** In a bag there are 4 blue pens and 6 black pens. What is P(Blue) in simplest form? **(2 marks)**

Show your working

Answer: _____

4 **Question 5:** A bar chart shows sandwich sales: Ham 15, Cheese 25, Tuna 10. What is the modal sandwich? **(2 marks)**

Show your working

Answer: _____

5 **Question 6:** A conversion graph shows £10 = 12 Euros. Convert £30 to Euros. **(2 marks)**

Show your working

Answer: _____

LEVEL 1 REVIEW • EXAM STYLE

CP3

Checkpoint Assessment 3 • Part 3/4

5 **Question 7:** A delivery depot records packages delivered: Mon 45, Tue 52, Wed 48, Thu 55, Fri 50. Calculate the mean daily deliveries. **(3 marks)**

Show your working

Answer: _____

6 **Question 8:** In a pie chart representing 120 people, the sector for 'Public Transport' has a right angle (90°). How many people take public transport? **(3 marks)**

Show your working

Answer: _____

7 **Question 9:** A pictogram uses a circle symbol ● = 8 cars. Monday shows 3 circles. Tuesday shows 2 1/2 circles. How many cars in total were washed over both days? **(3 marks)**

Show your working

Answer: _____

CP3

LEVEL 1 REVIEW • EXAM STYLE

Checkpoint Assessment 3 • Part 4/4

7 **Question 10:** Team Alpha mean score is 42, range is 5. Team Beta mean score is 42, range is 16. Write two comparison statements. **(3 marks)**

Show your working

Answer: _____

8 **Question 11:** A box holds 20 light bulbs: 2 are defective. What is the probability that a bulb chosen at random is NOT defective? Give your answer as a decimal. **(3 marks)**

Show your working

Answer: _____

9 **Question 12:** A survey asked 60 customers their preferred payment: Card (35), Cash (15), Phone App (10). What percentage preferred Card? **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

1 **Question 1:** Work out without a calculator: 418 × 14. Show your full column method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account balance is -£160. A wages payment of +£520 is paid in, and a bill of £180 is paid out. What is the new balance? **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

3 **Question 3:** Calculate without using a calculator: 1,638 ÷ 7. Show your bus stop division. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** Work out: 18 - 2 × 5 + 4. **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 3 • NO CALCULATOR ALLOWED

M1A

Mock Exam 1 • Section A (Non-Calculator)

5 **Question 5:** Simplify fully: 16/28. **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £150 in the ratio 3 : 2. What is the smaller share? **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

7 **Question 7:** A winter coat is priced at £75.00. It is discounted by 20% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

8 **Question 8:** A pancake recipe for 4 people requires 250 g flour and 300 ml milk. How much flour and milk are needed for 6 people? **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

9 **Question 9:** A delivery van travels 45 miles. Using 5 miles = 8 km, convert 45 miles into kilometres. **(3 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape gardener lays turf on a rectangular lawn measuring 8.5 m by 6.0 m. Turf rolls cover 1 m² each and cost £4.50. Delivery is £20.00. Calculate the total cost. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

11 **Question 11:** A healthcare assistant works 30 basic hours at £11.50/hr, plus 4 hours overtime at time-and-a-half (£17.25/hr). What is her total gross pay? **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A water cistern measures 1.5 m long, 0.8 m wide, and 0.5 m deep. How many litres of water does it hold when full? (1 m^3 = 1000 litres) **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 4 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

13 **Question 13:** A train departs Manchester at 10:48 and arrives in Leeds at 12:05. How long did the train journey take in hours and minutes? **(4 marks)**

Show your working

Answer: _____

14 **Question 14:** Daily parcel deliveries by a driver: Mon 38, Tue 45, Wed 42, Thu 49, Fri 46. Calculate: a) The mean daily deliveries b) The range. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 5 OF 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Allowed)

15 **Question 15:** On a scale drawing of a garden (1 : 50), a patio measures 12 cm by 8 cm. Calculate the real perimeter of the patio in metres, and the real area in square metres. **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A clinic audits 100 patients: 60 are female. 45 females had a flu vaccination. 25 males had a flu vaccination. Construct a two-way table. What is the probability that a patient chosen at random was vaccinated? **(5 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

1 **Question 1:** Work out without a calculator: 418 × 14. Show your full column method. **(3 marks)**

Show your working

Answer: _____

2 **Question 2:** A bank account balance is -£160. A wages payment of +£520 is paid in, and a bill of £180 is paid out. What is the new balance? **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

3 **Question 3:** Calculate without using a calculator: 1,638 ÷ 7. Show your bus stop division. **(3 marks)**

Show your working

Answer: _____

4 **Question 4:** Work out: 18 - 2 × 5 + 4. **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 3 • NO CALCULATOR ALLOWED

M2A

Mock Exam 2 • Section A (Non-Calculator)

5 **Question 5:** Simplify fully: 16/28. **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** Share £150 in the ratio 3 : 2. What is the smaller share? **(2 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 1 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

7 **Question 7:** A winter coat is priced at £75.00. It is discounted by 20% in a sale. Calculate the sale price. **(3 marks)**

Show your working

Answer: _____

8 **Question 8:** A pancake recipe for 4 people requires 250 g flour and 300 ml milk. How much flour and milk are needed for 6 people? **(3 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 2 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

9 **Question 9:** A delivery van travels 45 miles. Using 5 miles = 8 km, convert 45 miles into kilometres. **(3 marks)**

Show your working

Answer: _____

10 **Question 10:** A landscape gardener lays turf on a rectangular lawn measuring 8.5 m by 6.0 m. Turf rolls cover 1 m² each and cost £4.50. Delivery is £20.00. Calculate the total cost. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 3 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

11 **Question 11:** A healthcare assistant works 30 basic hours at £11.50/hr, plus 4 hours overtime at time-and-a-half (£17.25/hr). What is her total gross pay? **(4 marks)**

Show your working

Answer: _____

12 **Question 12:** A water cistern measures 1.5 m long, 0.8 m wide, and 0.5 m deep. How many litres of water does it hold when full? (1 m^3 = 1000 litres) **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 4 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

13 **Question 13:** A train departs Manchester at 10:48 and arrives in Leeds at 12:05. How long did the train journey take in hours and minutes? **(4 marks)**

Show your working

Answer: _____

14 **Question 14:** Daily parcel deliveries by a driver: Mon 38, Tue 45, Wed 42, Thu 49, Fri 46. Calculate: a) The mean daily deliveries b) The range. **(4 marks)**

Show your working

Answer: _____

OFFICIAL EXAM PRACTICE • PART 5 OF 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Allowed)

15 **Question 15:** On a scale drawing of a garden (1 : 50), a patio measures 12 cm by 8 cm. Calculate the real perimeter of the patio in metres, and the real area in square metres. **(5 marks)**

Show your working

Answer: _____

16 **Question 16:** A clinic audits 100 patients: 60 are female. 45 females had a flu vaccination. 25 males had a flu vaccination. Construct a two-way table. What is the probability that a patient chosen at random was vaccinated? **(5 marks)**

Show your working

Answer: _____

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX

Complete Step-by-Step Worked Solutions, Formulae & Glossary

TOPIC 1: PLACE VALUE & LARGE NUMBERS

Q1: 45,820; **Q2:** 128,450 - 95,200 = **33,250 units**; **Q3:** Order: 3,450; 12,890; 14,200; 108,500 [7m].

TOPIC 2: NEGATIVE NUMBERS IN PRACTICAL CONTEXTS

Q1: −11°C, −8°C, −2°C, 0°C, +4°C; **Q2:** −15 + 6 = **−9°C**; **Q3:** £120 − £185 = **−£65** (need £115 to reach +£50) [7m].

TOPIC 3: COLUMN ADDITION & SUBTRACTION WITH MONEY

Q1: 4,850 + 2,680 = **7,530**; **Q2:** £24.50 + £18.75 + £9.20 = **£52.45**; **Q3:** £1,500 − £845 − £320 = **£335.00** [7m].

TOPIC 4: LONG MULTIPLICATION IN CONTEXT

Q1: 145 × 18 = **2,610**; **Q2:** 24 crates × 35 items = **840 items**; **Q3:** 32 hours × £12.50 = **£400.00** [7m].

TOPIC 5: DIVISION WITH REMAINDERS IN CONTEXT

Q1: 256 ÷ 8 = **32**; **Q2:** 185 ÷ 12 = 15 remainder 5 → **16 crates needed**; **Q3:** £350 ÷ 4 = **£87.50 each** [7m].

TOPIC 6: FRACTIONS OF QUANTITIES

Q1: 1/4 of £240 = **£60.00**; **Q2:** 3/4 of 480 kg = (480 ÷ 4) × 3 = **360 kg**; **Q3:** 2/3 of 90 minutes = **60 minutes** [7m].

TOPIC 7: DECIMALS & ROUNDING

Q1: 14.86 to 1 d.p. = **14.9**; **Q2:** £3.45 + £8.80 = **£12.25**; **Q3:** 4.2 m - 1.85 m = **2.35 metres** [7m].

TOPIC 8: PERCENTAGES: 10%, 25%, 50% & 75%

Q1: 10% of £450 = **£45.00**; **Q2:** 25% discount on £180 = £45 → Sale price = **£135.00**; **Q3:** 75% of 320 staff = **240 staff** [7m].

TOPIC 9: RATIO SHARING IN 2 PARTS

Q1: Share £180 in ratio 2 : 1 → 3 parts. Shares: **£120 and £60**; **Q2:** Cement to sand 1 : 4. Total 25 kg → **5 kg cement, 20 kg sand**; **Q3:** Paint mix: **4 litres white, 2 litres blue** [7m].

TOPIC 10: PROPORTION & SCALING RECIPES

Q1: Soup for 4 is 600 ml → for 6 people = 150 × 6 = **900 ml**; **Q2:** 3 books cost £24 → 5 books = **£40.00**; **Q3:** Cake mix doubled: **500 g flour, 300 g sugar** [7m].

TOPIC 11: ESTIMATION & CHECKING

Q1: 49 × 21 ≈ 50 × 20 = **1,000**; **Q2:** 19 × £11.85 ≈ 20 × £12 = **£240.00**; **Q3:** Check subtraction by reverse addition: 316 + 184 = **500** [7m].

TOPIC 12: MONEY & GIVING CHANGE

Q1: Total = £3.45 + £2.80 + £6.15 = £12.40. Change from £20 = **£7.60**; **Q2:** £50 - £34.20 = **£15.80**; **Q3:** 3 items at £4.50 = £13.50. Change from £20 = **£6.50** [7m].

TOPIC 13: TIME: 12H & 24H CLOCK

Q1: 3:15 pm = **15:15**; 19:40 = **7:40 pm**; **Q2:** 08:30 to 16:45 = **8 hours 15 minutes**; **Q3:** Meeting starts 14:30, lasts 1h 45m → finishes **16:15** [7m].

TOPIC 14: TIMETABLES & JOURNEY PLANNING

Q1: 14:22 to 14:58 = **36 minutes**; **Q2:** Next train after 10:15 is at 10:32, arrives **11:05**; **Q3:** Journey time: **1 hour 18 minutes** [7m].

TOPIC 15: METRIC LENGTH: MM, CM, M, KM

Q1: 2.4 m = **240 cm**; 45 mm = **4.5 cm**; **Q2:** 3.6 m - (4 × 0.65 m) = **1.0 metre remaining**; **Q3:** Total running: 5.2 km = **5,200 metres** [7m].

TOPIC 16: METRIC WEIGHT: G AND KG

Q1: 1,750 g = **1.75 kg**; **Q2:** 4 parcels at 350 g = 1,400 g = **1.4 kg**; **Q3:** Flour: 2.5 kg - 800 g = **1.7 kg** (1,700 g) [7m].

TOPIC 17: METRIC CAPACITY: ML AND LITRES

Q1: 4.5 litres = **4,500 ml**; **Q2:** 15 × 250 ml = 3,750 ml = **3.75 litres**; **Q3:** Dilution: 500 ml concentrate into 4.5 L water = **5.0 litres total** [7m].

TOPIC 18: PERIMETER OF RECTANGLES

Q1: 2(9 + 6) = **30 metres**; **Q2:** Room 4.5 m × 3.2 m → P = 2(7.7) = **15.4 m**; **Q3:** Fencing needed = **48 metres** [7m].

TOPIC 19: AREA OF RECTANGLES

Q1: 4.5 × 3 = **13.5 m²**; **Q2:** Wall 6 × 2.5 = **15 m²** (2 coats = **30 m²**); **Q3:** 8 × 5 = **40 m²**, Cost = 40 × £12 = **£480.00** [7m].

TOPIC 20: 2D & 3D SHAPES & ANGLES

Q1: Shape with 6 rectangular faces is a **cuboid**; **Q2:** Right angle = **90°**; **Q3:** Triangular prism has **5 faces, 9 edges, 6 vertices** [7m].

TOPIC 21: COMPASS DIRECTIONS & HEADINGS

Q1: Facing North, turn 90° clockwise → **East**; **Q2:** Opposite of South-West is **North-East**; **Q3:** Facing East, turn 180° → **West** [7m].

TOPIC 22: TALLY CHARTS & FREQUENCY

Q1: 14 tallies: 2 bundles of 5 and 4 single lines; **Q2:** Frequency total = 8 + 14 + 12 + 6 = **40**; **Q3:** Most popular category is **Van** (14 tallies) [7m].

TOPIC 23: BAR CHARTS: SCALES & COMPARISON

Q1: Wed = 40, Mon = 25 → Difference = **15 sales**; **Q2:** Scale: 1 square = 5 units; **Q3:** Total weekly sales = **165 units** [7m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

TOPIC 24: PICTOGRAMS WITH KEY SYMBOLS

Q1: 1 circle = 8 cars. 3.5 circles = 3 × 8 + 4 = **28 cars**; **Q2:** 20 bicycles with key = 4 → **5 symbols**; **Q3:** Total vehicles = **72 vehicles** [7m].

TOPIC 25: MEAN AVERAGE

Q1: Daily hours: 7 + 8 + 8 + 9 + 8 = 40 → 40 ÷ 5 = **8 hours**; **Q2:** Scores: 15, 18, 21, 26 → 80 ÷ 4 = **20**; **Q3:** Mean weekly spend = **£42.50** [7m].

TOPIC 26: RANGE: MEASURE OF SPREAD

Q1: Calls: 71 - 38 = **33 calls**; **Q2:** Temperatures: +6 - (-4) = **10°C**; **Q3:** Range of delivery times = 45 - 18 = **27 minutes** [7m].

TOPIC 27: READING INFORMATION FROM TABLES

Q1: 3 small parcels at £3.29 = **£9.87**; **Q2:** Table lookup: Standard delivery = **£4.50**; **Q3:** Invoice total: **£148.20** [7m].

TOPIC 28: PROBABILITY WORDS

Q1: Rolling a number 1 to 6 on standard die is **Certain**; **Q2:** Flipping tails on a coin is **Evens** (50-50); **Q3:** Rolling an 8 on standard die is **Impossible** [7m].

CHECKPOINT 1: NUMBER & NUMBER SYSTEMS (STRAND 1)

Q1: 324 × 15 = **4,860** [2m]. **Q2:** £45.00 - £28.40 = **£16.60** [2m]. **Q3:** 3/4 of £180 = **£135.00** [2m]. **Q4:** 148 ÷ 12 = 12 r 4 → **13 boxes needed** [2m]. **Q5:** 10% of £360 = **£36.00** [2m]. **Q6:** Share £150 ratio 3:2 → **£90 and £60** [3m]. **Q7:** Estimate: 48 × 19 ≈ 50 × 20 = **1,000** [2m]. **Q8:** 25% discount on £84 = £21 → **£63.00** [3m]. **Q9:** 156 × 24 = **3,744** [2m]. **Q10:** Balance = £450 - £185 - £95 = **£170.00** [2m]. **Q11:** Proportion: 4 cakes need 500 g → 6 cakes = **750 g** [2m]. **Q12:** Check by reverse: 170 + 280 = **450** [3m].

CHECKPOINT 2: MEASURES, SHAPE & SPACE (STRAND 2)

Q1: 2(8 + 5) = **26 metres** [2m]. **Q2:** Area = 7.5 × 4 = **30 m²** [2m]. **Q3:** 3.5 m = **350 cm** [2m]. **Q4:** 14:35 to 16:15 = **1 hour 40 minutes** [2m]. **Q5:** Change from £20 for bill £14.25 = **£5.75** [2m]. **Q6:** Weight: 2.4 kg = **2,400 g** [3m]. **Q7:** Capacity: 4 × 750 ml = 3,000 ml = **3 litres** [2m]. **Q8:** Solid with 6 square faces is a **cube** [2m]. **Q9:** Facing West, turn 90° clockwise → **North** [2m]. **Q10:** Perimeter of garden: **34 metres** [2m]. **Q11:** Carpet cost: 24 m² × £15 = **£360.00** [3m]. **Q12:** 24h clock: 8:45 pm = **20:45** [2m].

CHECKPOINT 3: HANDLING INFORMATION & DATA (STRAND 3)

Q1: Frequency total = 12 + 18 + 15 + 5 = **50** [2m]. **Q2:** Mean = (6 + 8 + 9 + 7 + 10) ÷ 5 = 40 ÷ 5 = **8** [2m]. **Q3:** Range = 10 - 6 = **4** [2m]. **Q4:** Pictogram: 1 star = 4 sales. 4.5 stars = **18 sales** [2m]. **Q5:** Bar chart: Category B has highest count = **35** [2m]. **Q6:** Probability of rolling odd number on die = **Evens** (3/6) [2m]. **Q7:** Table lookup: Delivery charge for 5 kg = **£6.50** [2m]. **Q8:** Mean score: **74%** [2m]. **Q9:** Difference in sales between Mon and Fri = **22 units** [2m]. **Q10:** Probability of choosing red card from standard deck = **Evens** [2m]. **Q11:** Comparison: Team A is more consistent (range 6 vs 15) [3m]. **Q12:** Survey total = **120 respondents** [2m].

OFFICIAL MOCK EXAM 1 (56 MARKS) • SECTION A

Section A (Non-Calculator, 14 Marks):
Q1: 348 × 16 = **5,568** [M1 column, A1: 2m].
Q2: £50.00 - £31.65 = **£18.35** [M1 sub, A1: 2m].
Q3: 1/4 of £360 = **£90.00** [M1 div, A1: 2m].
Q4: 195 ÷ 15 = **13** [M1 bus stop, A1: 2m].
Q5: 10% of £480 = **£48.00** [M1, A1: 2m].
Q6: 2(12 + 8) = **40 metres** [M1 formula, A1, B1 check: 4m].

OFFICIAL MOCK EXAM 1 (56 MARKS) • SECTION B

Section B (Calculator Allowed, 42 Marks):
Q7: Room area = 6.5 × 4.2 = 27.3 m². Carpet cost = 27.3 × £18 = **£491.40** [P1, M1, A1, B1: 4m].
Q8: 25% discount on £320 = £80 → Sale price = **£240.00** [P1, M1, A1, B1: 4m].
Q9: Shift = 08:45 to 17:15 = 8h 30m. Minus 45m break = **7 hours 45 minutes** [P1, M1, A1, B1: 4m].
Q10: 37.5 hrs × £11.80 = **£442.50** [P1, M1, A1, B1: 4m].
Q11: Mean = (24 + 31 + 28 + 35 + 32) ÷ 5 = 150 ÷ 5 = **30 calls** [P1, M1, A1, B1: 4m].
Q12: Range = 35 - 24 = **11 calls** [P1, M1, A1, B1: 4m].
Q13: Ratio 3:1 → 4 parts. £400 ÷ 4 = £100. Shares: **£300 and £100** [P1, M1, A1, B1: 5m].
Q14: Volume = 4 × 2.5 × 1.8 = 18 m³ = **18,000 litres** [P1, M1, A1, B1: 5m].
Q15: Travel time = 165 ÷ 55 = **3 hours** [P1, M1, A1: 4m].
Q16: Full comparison showing Team B has higher mean (28 vs 24) and greater consistency [P1, M1, A1, B1: 4m].

OFFICIAL MOCK EXAM 2 (56 MARKS) • SECTION A

Section A (Non-Calculator, 14 Marks):
Q1: 425 × 18 = **7,650** [2m]. **Q2:** £100 - £64.35 = **£35.65** [2m]. **Q3:** 3/4 of £240 = **£180.00** [2m]. **Q4:** 288 ÷ 12 = **24** [2m]. **Q5:** 50% of £190 = **£95.00** [2m]. **Q6:** Perimeter = 2(15 + 9) = **48 m** [4m].

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

OFFICIAL MOCK EXAM 2 (56 MARKS) • SECTION B

Section B (Calculator Allowed, 42 Marks):
Q7: Paving area = 8 × 5 = 40 m². Total cost = 40 × £22.50 = **£900.00** [4m]. **Q8:** 15% discount on £420 = £63 → **£357.00** [4m]. **Q9:** Train departs 11:24, arrives 13:08 → **1 hour 44 minutes** [4m]. **Q10:** Overtime: (35 × 11.50) + (5 × 17.25) = **£488.75** [4m]. **Q11:** Mean daily sales = 180 ÷ 5 = **36 units** [4m]. **Q12:** Range = 45 - 22 = **23 units** [4m]. **Q13:** Cement mix: 1:4 ratio for 50 kg → **10 kg cement, 40 kg sand** [5m]. **Q14:** Capacity = 2.5 × 1.2 × 1.0 = 3 m³ = **3,000 litres** [5m]. **Q15:** Average speed = 120 ÷ 2 = **60 mph** [4m]. **Q16:** Decision: Supplier A is cheaper by £45.00 [4m].

STATUTORY LEVEL 1 FORMULAE & CONVERSIONS

Perimeter & Area of Rectangles:

• Perimeter = 2 × (Length + Width) (or sum of all 4 sides)

• Area = Length × Width (answer in m², cm², or mm²)

Fractions, Decimals & Percentages:

• Fraction of Amount = (Amount ÷ Denominator) × Numerator

• 50% = 1/2 (divide by 2) 25% = 1/4 (divide by 4)

• 10% = 1/10 (divide by 10) 75% = 3/4 (3 × 25%)

Averages & Data:

• Mean = Sum of all values ÷ Total number of values

• Range = Highest value − Lowest value

Metric Conversions:

• 1 km = 1,000 m 1 m = 100 cm 1 cm = 10 mm

• 1 kg = 1,000 g 1 litre = 1,000 ml

MATHEMATICAL GLOSSARY (PART 1)

Area
The amount of flat space inside a 2D shape, measured in square units (l × w).

Bar Chart
A diagram using rectangular bars of equal width to compare categories.

Decimal
A number containing a point separating whole units from tenths and hundredths.

Denominator
The bottom number of a fraction showing how many equal parts make the whole.

Estimation
Rounding numbers to simplify a calculation and check if an answer is sensible.

Frequency
The number of times a particular event or category occurs.

MATHEMATICAL GLOSSARY (PART 2)

Mean
The average found by adding all values and dividing by the total count.

Numerator
The top number of a fraction showing how many parts are being considered.

Perimeter
The total distance all the way around the outside edge of a shape.

Pictogram
A chart that uses pictures or symbols to represent data, explained by a key.

Range
The difference between the highest and lowest values in a data set.

Ratio
A comparison of two or more quantities showing their relative sizes (e.g. 2 : 1).

REFERENCE SECTION: SOLUTIONS, FORMULAE & INDEX (CONTINUED)

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Candidate Calculation Notes & Formula Log

AUDIT

FINAL REVIEW • EXAM READINESS

Candidate Self-Audit & Progress Matrix

Core Functional Skills Level 1 Mathematics Competencies

Check off your skills before taking your Functional Skills Maths Level 1 exam:

- I can add, subtract, multiply, and divide whole numbers and decimals.
- I can calculate simple fractions (1/2, 1/4, 3/4) and percentages (10%, 25%, 50%).
- I can convert between metric units (mm, cm, m, km; g, kg; ml, l).
- I can calculate perimeter and area of rectangular spaces.
- I can read information from tables, bar charts, and timetables.
- I always show my working and write the correct units (£, m, kg).

✎ Personal Exam Day Checklist & Targets

Build solid numerical confidence and pass Level 1 Maths

Tailored specifically for Level 1 candidates. Breaks down daunting arithmetic into friendly, visual steps: place value, times tables, decimals, fractions of amounts, simple percentages, metric unit conversions, perimeter, area, and bar charts.

INSIDE THIS PUBLICATION

- ✓ **Visual, Dyslexia-Friendly Layout:** Clear diagrams, color-coded arithmetic steps, and simplified chunked explanations.
- ✓ **Non-Calculator Mental Strategies:** Multiplication grids, division remainders, and fraction models that demystify Section A.
- ✓ **Everyday Practical Contexts:** Real-world problems: budgeting wages, train timetables, shopping deals, tiling and fencing.
- ✓ **Clear Step-by-Step Exemplars:** Over 50 illustrated model answers demonstrating how to lay out workings cleanly.
- ✓ **Key Formulas & Conversions:** Quick recall conversion charts for length (mm/cm/m/km), weight (g/kg), and time.

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